

Introduction

An index is simply a structured database that is up-to-date by a search engine. It consists of page content, links, metadata, and other valuable information that helps in indexing and locating the webpages as a query given by the user. The main purpose of the indexes is to help search engines. Like we use in Google and Yahoo searches. It takes the user query or request and replies promptly. Search engines crawl the web pages to find the new pages and search for updated ones. This is known as the crawling process.

In the information retrieval (IR) system, it has control over the corpus and vocabulary, whereas in web search, it goes on the whole web, so there is no control over the corpus. There is a power method to solve the page rank. Pages are indexed based on various factors like keywords, relevance, and quality. It gives accurate and relevant results to the users.



Let us assume that the number of pages in index A is $|A|$ and the number of pages in index B is $|B|$. It is said that 30% of the A pages are available in the B index. 50% of B pages are available in the A index.

Mathematically, it is represented as below:

The percentage of A's pages in B's index is 30% of $|A|$.

The percentage of B's pages in A's index is 50% of $|B|$.

Here, to be consistent in both ways, there must be an equal number of A pages in the B index and B pages in the A index.

$$30\% \text{ of } |A| = 50\% \text{ of } |B|$$

$$|A| = 1.66 |B|$$

It means that the A index is 1.66 times larger than B's index.

It helps in a variety of ways and is very important to know the flow, as discussed below:

It provides the allocation of resources for storing, indexing, and crawling based on the size of the indexes. It helps in the relevance search. It enhances the search engines improving.